

Project Scope — ResViz - West Coast Vikings

1. Goal

ResViz will replace the Learner Admissions committee's manual process for determining satisfiability of propositional logic knowledge bases with a web application that parses, resolves, and visually shows how each conclusion was reached; this is so that committee members can see and understand the reasoning behind a result.

2. In-scope — backlog items, in build order

High priority

- Backlog 1 — UI: create a project, input/upload a knowledge base, enter a proposition via keyboard or symbol palette, view the resolution output.
- Backlog 2 — Knowledge base parser: converts an uploaded knowledge base into clause form.
- Backlog 3 — CNF converter: converts clause form into CNF for both the knowledge base and the entered proposition.
- Backlog 7 — Reusable API: connects front and back end, built to be reusable

Medium priority:

- Backlog 5 — Resolution algorithm: priority-based resolution, checking clauses most closely related to the entered proposition first rather than blind pairwise comparison.

Low priority:

- Backlog 4 — Resolution step visualiser: step-forward/backward through each clause comparison like a debugger, disproving comparison highlighted in red, ending in the conclusion.
- Backlog 6 — Enhance UX: polish pass — animation, transitions, colour palette, intuitive flow.

3. Out of scope

- First-order logic — propositional logic only.
- User accounts / persistent multi-user login — single-session project working, not a multi-user platform. The project dies with the session.
- Mobile-native support – this is a web application only.
- Optimizing algorithm speed and performance

4. Assumptions

- No agreed domain-specific language exists for writing facts; the team will define its own syntax for the parser or use the syntax that works best with the given parser.
- No agreed visualization approach exists; the team will set its own convention, reviewed with the client via a prototype/mock-up.
- The team will reuse and rewrite the existing client-provided Python parser, CNF converter, and resolution algorithm as a starting point.

5. Traceability

Client need	Backlog items
Quick typing of formulas	1
Ease of annotation	Pending — not yet in backlog
Trace/explain how the answer was reached	4
Prioritise clauses related to goal, not blind comparison	5
Split backend/frontend, reusable API	7
Supporting infrastructure	2, 3